A Hacker’s Guide to Speculative Decoding in vLLM

CUDA MODE talk by Cade Daniel

Introductions

- Working on LLM inference in vLLM
- Software Engineer at Anyscale
- Previously, model parallelism systems at AWS
  - https://arxiv.org/abs/2111.05972
- Feel free to reach out!
  - https://x.com/cdnamz

Topics

- vLLM’s core principles
- Spec decode background
- Spec decode framework intro (for contributors)
- Future contribution ideas
- Q&A (30min)

vLLM’s core principles

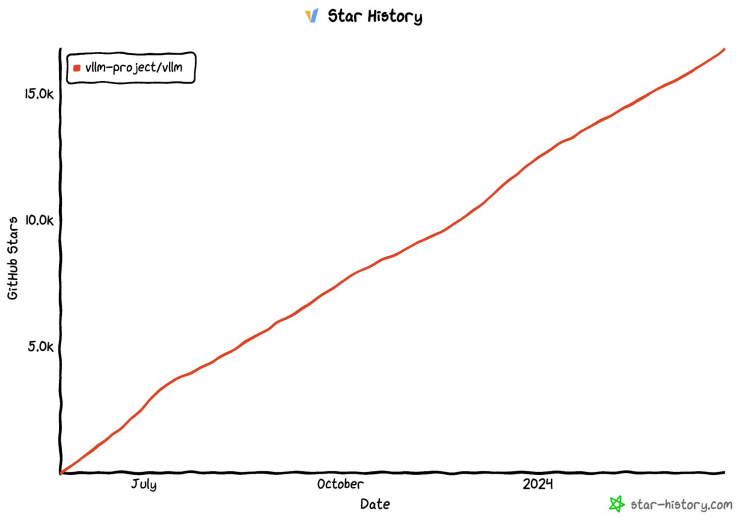
- Ease-of-use
- Great performance
- Hardware agnosticity

Easy-to-use

https://github.com/vllm-project/vllm

```
$ pip install vllm
```

17K Stars



Great performance

Performance features

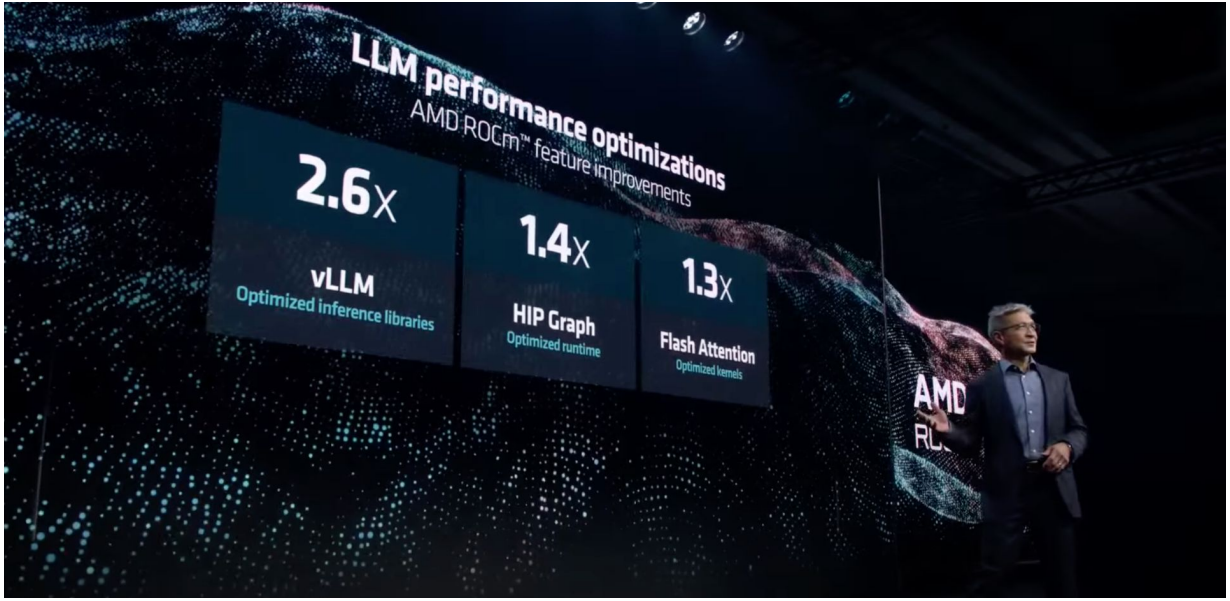
- PagedAttention/tensor parallelism
- Optimized multi-LoRA
- Chunked prefill
- Automatic prefix caching
- Guided decoding
- Quantization (fp8 WIP, and others)
- Pipeline-parallelism (WIP)
- Prefill disaggregation (WIP)

More contributions welcome!

Hardware agnosticity

Current backends:

- NVIDIA, AMD, Inferentia, TPU (WIP), CPU



Source: [AMD Presents: Advancing AI]

Topics

- vLLM’s core principles
- **Spec decode background**
- Spec decode framework intro (for contributors)
- Future contribution ideas
- Q&A (30min)

Spec decode background

- Recommended reading: Andrej Karpathy's tweet on speculative decoding
  - https://x.com/karpathy/status/1697318534555336961
- Memory-boundedness
  - In memory-bound LLM inference, the full GPU compute capacity is underutilized
  - The unused compute can be used, if we can find a way to use it
- Not all parameters required for every token
  - Do we really need 70B parameters to answer "What is the capital of California"? Probably not…
- Idea:
  - Try to predict what large model will say
  - Get probabilities of predictions
  - Use heuristic to accept or rejection the predictions based on probabilities

Spec decode background

```
def below_threshold(l: list, t: int):
    """Return True if all numbers in the list l are below threshold t.
    >>> below_threshold([1, 2, 4, 10], 100)
    True
    >>> below_threshold([1, 20, 4, 10], 5)
    False
    """
    if isinstance(l, list):
        return True
    else:
        if t <= l < below_threshold(l, t):
            return True
    else:
        # If the first l element of l is an integer, then it is
        # the whole range of integers.
        if not isinstance(l[0], list):
            return True
    else:
        # If the first l element of l is a str, then it is
        # the whole string.
        if hasattr(l, 'findlen'):
            return findlen(l)
        return False


def thresh(t: int, max: int) -> int:
    """Return
```

Figure 3: A visualization of the origin of tokens in an example T=1 HumanEval completion. Green background originates with the N-gram draft[2] model, blue the draft model, and red the oracle model. (Of course, all tokens are eventually checked by the oracle model.) Obvious tokens – like whitespace – are preferentially accelerated relative to difficult ones.

Source: “Accelerating LLM Inference with Staged Speculative Decoding” https://arxiv.org/pdf/2308.04623

How to evaluate speedup?

- Recommended reading: “Fast Inference from Transformers via Speculative Decoding” https://arxiv.org/pdf/2211.17192

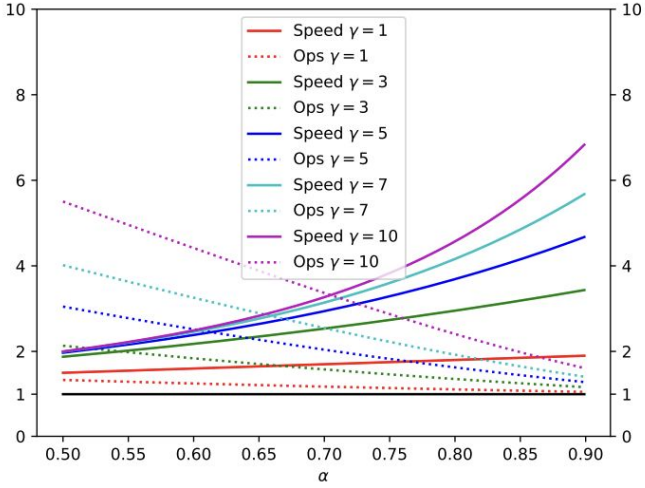


Figure 4. The speedup factor and the increase in number of arithmetic operations as a function of α for various values of γ.

How to evaluate speedup?

- Simplified version:
  - Inter-token latency = step time / number of tokens per step in expectation
  - Example without speculative decoding: 30ms / 1 → 1 token per 30ms
  - Example with speculative decoding: 40ms / 2.5 → 1 token per 16ms
- Key factors
  - How long does it take to propose?
  - How accurate are the proposals?
  - How long does it take to verify / other spec framework overheads?
- In practice:
  - https://github.com/vllm-project/vllm/blob/main/vllm/spec_decode/metrics.py
  - Acceptance rate – “How aligned is the proposal method with the target model?”
  - System efficiency – “How efficient is the deployment compared to 100% acceptance rate?”

Losslessness

- Is the output of speculative decoding different than the target model?
  - TL;DR No if using rejection sampling, subject to hardware numerics
  - Diagram https://github.com/vllm-project/vllm/pull/2336
  - Yes if using lossly sampling technique, e.g. Medusa's typical acceptance (but higher acceptance rate!)
- Recommended reading (proof of losslessness): Accelerating Large Language Model Decoding with Speculative Sampling

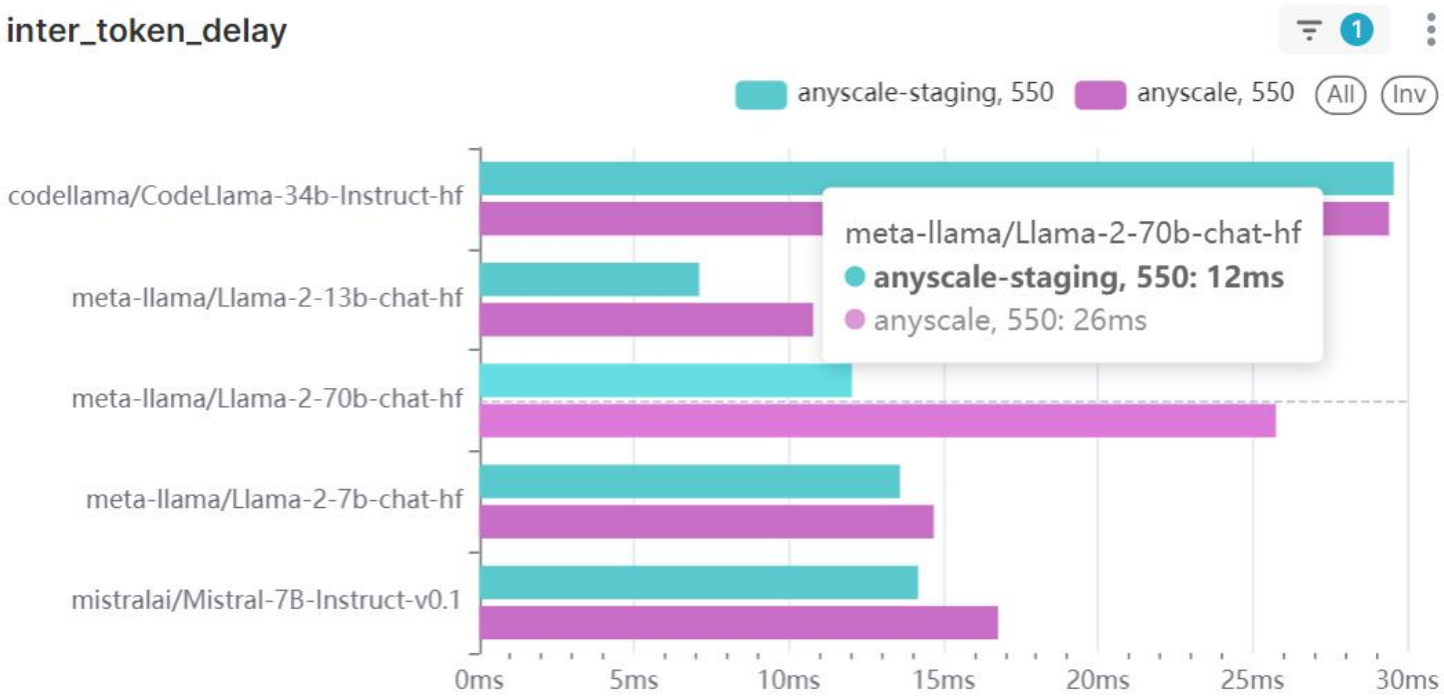
Topics

- vLLM’s core principles
- Spec decode background
- **Spec decode framework intro (for contributors)**
- Future contribution ideas
- Q&A (30min)

Current status in vLLM

- Spec decode framework is complete with correctness tests
- Supports draft model, ngram, Medusa (soon), IBM's MLPSpeculator (soon)
  - Other features like skipping speculation for some sequences, dynamic speculative decoding
- Missing performance optimizations to achieve Anyscale's internal fork performance
  - https://github.com/vllm-project/vllm/issues/4630
  - Llama2 70B 50% ITL reduction on BS=1..8 with temperature 1.0

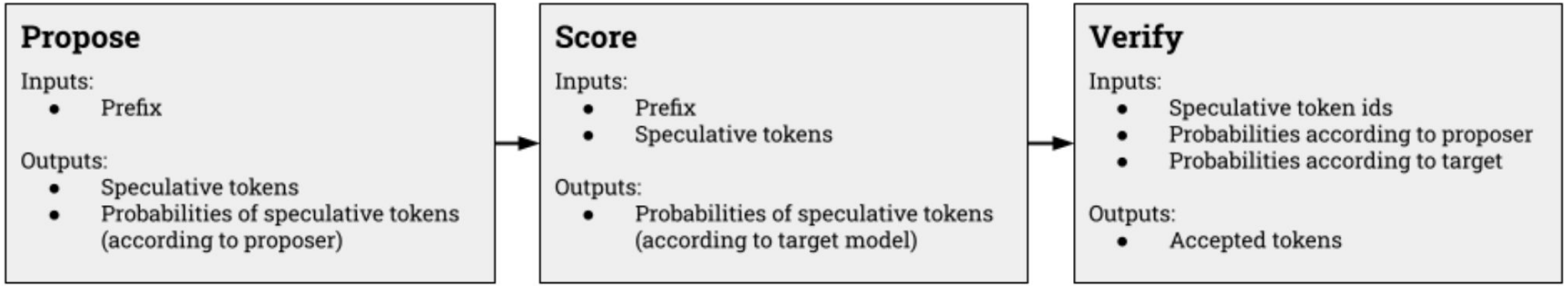
Current status in vLLM



Spec decode framework

- SpecDecodeWorker
  - Proposers (ngram, draft model)
  - Scorer (top-1 scoring)
  - Verifier (rejection sampling)

Spec decode framework



Proposer implementations	Verification implementations
• Draft model (staged, tree, cascading) • Medusa / EAGLE • BiTA (prompt tuning) • n-gram Jacobi (Lookahead) • Input-grounded speculation • RAG-grounded speculation • Note: multiple proposers can be combined, e.g. medusa + RAG-grounded)	• Lossless rejection sampling • Lossy rejection sampling • “Typical acceptance” (lossy medusa) • Greedy acceptance

Spec decode framework: links

- SpecDecodeWorker
  - https://github.com/vllm-project/vllm/blob/37464a0f745a0204da7443d2a6ef4b8f65e5af12/vllm/spec_decode/spec_decode_worker.py#L1
- Proposers
  - Draft model proposer https://github.com/vllm-project/vllm/blob/37464a0f745a0204da7443d2a6ef4b8f65e5af12/vllm/spec_decode/multi_step_worker.py#L1
  - Ngram proposer https://github.com/vllm-project/vllm/blob/37464a0f745a0204da7443d2a6ef4b8f65e5af12/vllm/spec_decode/ngram_worker.py#L1
- Verifier
  - Rejection sampler https://github.com/vllm-project/vllm/blob/37464a0f745a0204da7443d2a6ef4b8f65e5af12/vllm/model_executor/layers/rejection_sampler.py#L1
  - WIP typical acceptance https://github.com/vllm-project/vllm/issues/5015

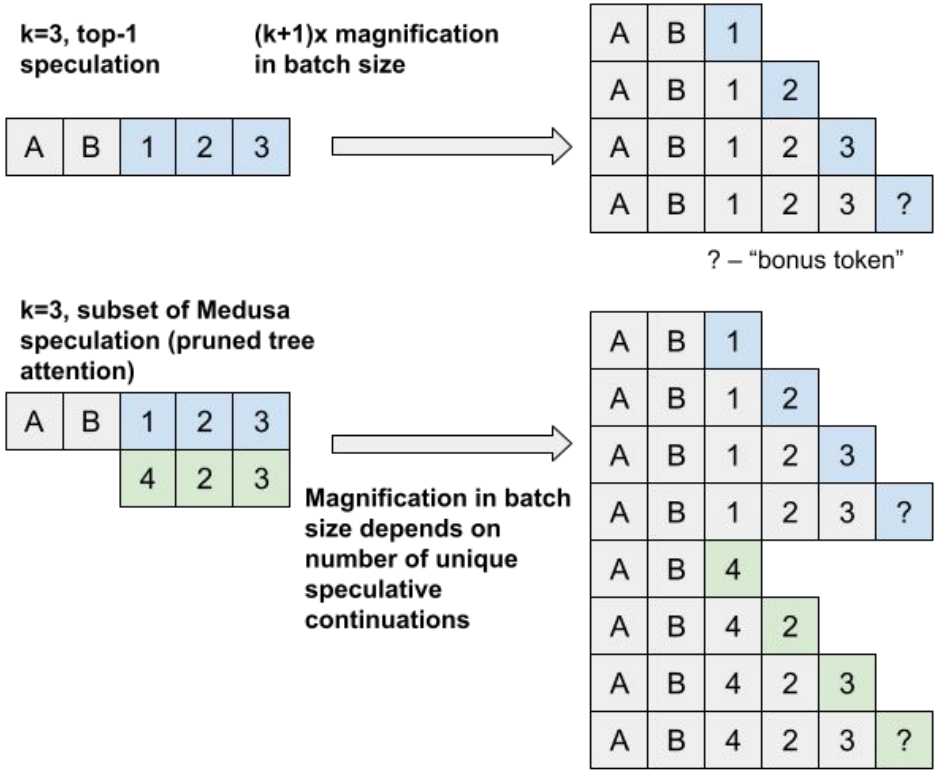
Spec decode framework: top1 vs top-k “tree attention”

- Top-1: proposal method suggests 1 token per sequence per slot
- Top-k: proposal method suggests k tokens per sequence per slot
- Recommended reading
  - https://sites.google.com/view/medusa-llm
  - https://arxiv.org/pdf/2305.09781
  - https://www.together.ai/blog/sequoia
- Currently only top-1 proposal and scoring is supported
  - Top-k is a future work
  - Most aggressive speedups require top-k attention masking
  - FlashInfer going to support masking
  - https://github.com/vllm-project/vllm/issues/3960

Spec decode framework: “Bonus token” and “recovered token”

- Recommended reading: https://arxiv.org/pdf/2302.01318
- Bonus token: All speculative tokens may be accepted. We can sample from target model distribution normally in this case
  - → we get an additional token in the happy-path!
- Recovered token: If all tokens are rejected, we can use math trick to sample a correct token from the target model distribution
  - → We always get >=1 token
- Logic is in rejection sampler / SpecDecodeWorker
  - https://github.com/vllm-project/vllm/blob/37464a0f745a0204da7443d2a6ef4b8f65e5af12/vllm/model_executor/layers/rejection_sampler.py#L210

Spec decode framework: “Bonus token” and “recovered token”



Spec decode framework: “Bonus token” and “recovered token”

- Recommended reading: https://arxiv.org/pdf/2302.01318
- Bonus token: All speculative tokens may be accepted. We can sample from target model distribution normally in this case
  - → we get an additional token in the happy-path!
- Recovered token: If all tokens are rejected, we can use math trick to sample a correct token from the target model distribution
  - → We always get >=1 token
- Logic is in rejection sampler / SpecDecodeWorker
  - https://github.com/vllm-project/vllm/blob/37464a0f745a0204da7443d2a6ef4b8f65e5af12/vllm/model_executor/layers/rejection_sampler.py#L210

Spec decode framework: “Bonus token” and “recovered token”

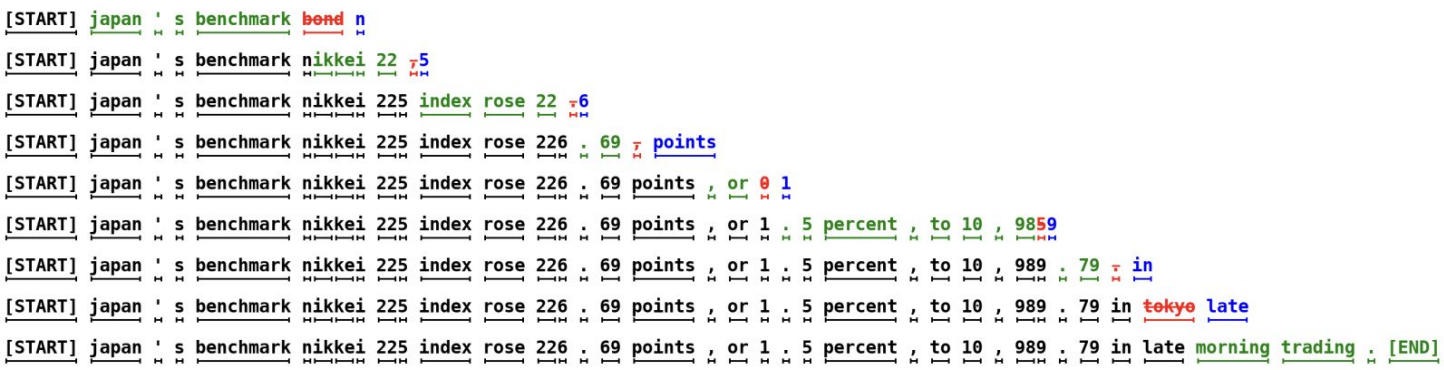


Figure 1. Our technique illustrated in the case of unconditional language modeling. Each line represents one iteration of the algorithm. The **green** tokens are the suggestions made by the approximation model (here, a GPT-like Transformer decoder with 6M parameters trained on lm1b with 8k tokens) that the target model (here, a GPT-like Transformer decoder with 97M parameters in the same setting) accepted, while the **red** and **blue** tokens are the rejected suggestions and their corrections, respectively. For example, in the first line the target model was run only once, and 5 tokens were generated.

https://arxiv.org/pdf/2211.17192

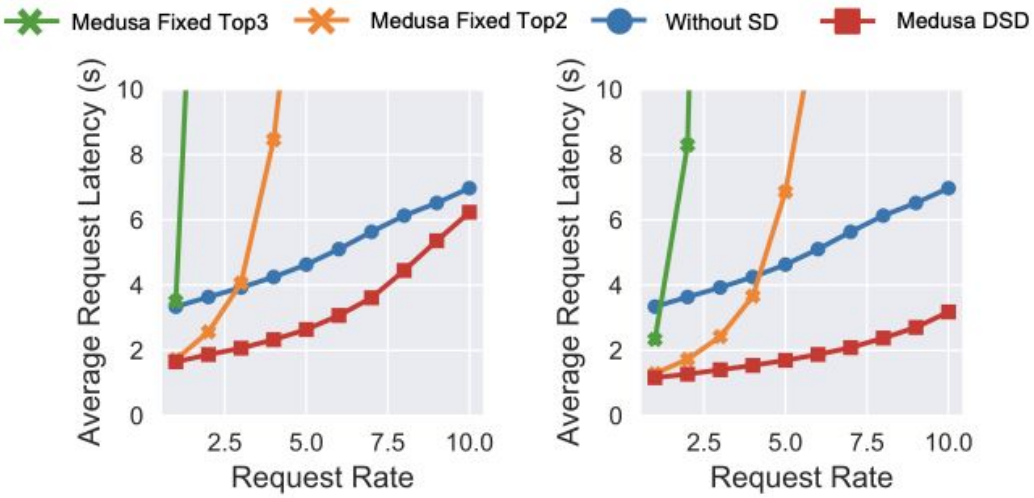
Lookahead scheduling

- Problem: Scoring speculative tokens generates KV. How can we save accepted KV to skip regeneration and reduce FLOPs requirements?
- Recommended reading: [What is lookahead scheduling in vLLM?]
- TL;DR:
  - vLLM's scheduler allocates additional space for KV
  - The SpecDecodeWorker uses the space to store KV of speculative tokens
  - Accepted token KV is stored correctly

Dynamic speculative decoding

- Problem: As batch size increases, spare FLOPs is reduced. How can we ensure spec decode performs no worse than no spec decode?
- Recommended reading: https://github.com/vllm-project/vllm/issues/4565
  - Work by Lily Liu and Cody Yu
- TL;DR
  - Based on the batch size, adjust which sequences have speculations (or disable spec dec altogether)
  - Future work: per-sequence speculation length

Dynamic speculative decoding



(a) $\alpha = 0.6$ **(b)** $\alpha = 0.8$

Source: Xiaoxuan Liu

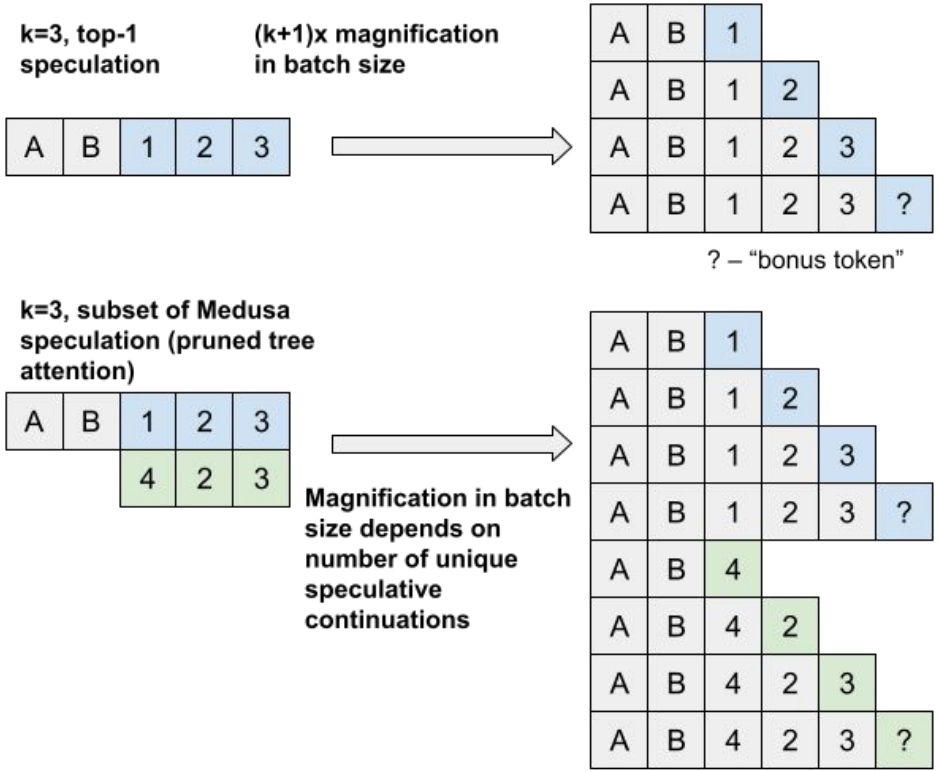
Dynamic speculative decoding

- Problem: As batch size increases, spare FLOPs is reduced. How can we ensure spec decode performs no worse than no spec decode?
- Recommended reading: https://github.com/vllm-project/vllm/issues/4565
  - Work by Lily Liu and Cody Yu
- TL;DR
  - Based on the batch size, adjust which sequences have speculations (or disable spec dec altogether)
  - Future work: per-sequence speculation length

Batch expansion

- Problem: How to support scoring when PagedAttention only supports 1 query token per sequence?
- Recommended reading: Optimizing attention for spec decode can reduce latency / increase throughput
- TL;DR
  - We create “virtual sequences” in SpecDecodeWorker each with 1 query token
  - This expands the batch (and duplicates KV loads in the attention layers)
  - We can remove this with an attention kernel which supports PagedAttention + multiple query tokens per sequence
  - Contact https://github.com/LiuXiaoxuanPKU for more information

Batch expansion



Batch expansion

- Problem: How to support scoring when PagedAttention only supports 1 query token per sequence?
- Recommended reading: Optimizing attention for spec decode can reduce latency / increase throughput
- TL;DR
  - We create “virtual sequences” in SpecDecodeWorker each with 1 query token
  - This expands the batch (and duplicates KV loads in the attention layers)
  - We can remove this with an attention kernel which supports PagedAttention + multiple query tokens per sequence
  - Contact https://github.com/LiuXiaoxuanPKU for more information

Testing

- Problem: How can we validate correctness of spec decode?
- TL;DR:
  - E2E: When temperature==0, we expect equality with and without spec decode
  - Rejection sampler unit tests (output distribution does not change regardless of draft/target probabilities))
- You can rely on these tests when contributing
  - https://github.com/vllm-project/vllm/tree/main/tests/spec_decode
  - https://github.com/vllm-project/vllm/tree/main/tests/spec_decode/e2e

Topics

- vLLM’s core principles
- Spec decode background
- Spec decode framework intro (for contributors)
- **Future contribution ideas**
- Q&A (30min)

Future contribution ideas

- More engineering
  - Retrieval-acceleration https://arxiv.org/html/2401.14021v1
  - Chunked prefill + spec decode
  - Prefix caching + spec decode
  - Guided decoding + spec decode
  - Inferentia / TPU / CPU support
- More modeling
  - Meta-model for speculation length
  - Meta-model for speculation type
- Large / mixed engineering+modeling
  - Multi-LoRA draft model (specialize to domains)
  - Online learning draft model https://arxiv.org/abs/2310.07177
  - Batched parallel decoding https://github.com/vllm-project/vllm/issues/4303

Thank you!

- Many people contributed
  - Lily Liu, Cody Yu, Antoni Baum, vLLM creators, Vikas Ummadisetty, Chen Shen, Sang Cho
  - Many others

Topics

- vLLM’s core principles
- Spec decode background
- Spec decode framework intro (for contributors)
- Future contribution ideas
- **Q&A (30min)**